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Network meta-analysis on the efficacy and safety of management for resectable stage IIIA-N2 non-small cell lung cancer

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Abstract

Background There is controversy regarding the optimal treatment for stage IIIA-N2 non-small cell lung cancer (NSCLC). We aimed to address this crucial issue through a frequentist network meta-analysis.

Methods We conducted a literature database search for randomized controlled trials comparing the following treatment modalities before March 1st, 2023: surgery, radiotherapy, chemotherapy, targeted therapy, immunotherapy, and various combinations of these treatments. Summary data on overall survival (OS) and treatment-related deaths (trDeath) were analyzed using frequentist methods.

Results Twenty-two randomized controlled trials (RCTs) with 3269 participants were included, covering 17 treatment regimens. In terms of overall survival, surgery followed by adjuvant targeted therapy (S-T), neoadjuvant targeted therapy followed by surgery and adjuvant targeted therapy (T-S-T), and neoadjuvant chemotherapy followed by surgery and adjuvant chemotherapy (C-S-C) were relatively more advantageous than other treatment regimens. Overall, S-T is the most likely treatment option to prolong OS, with a 59.8% likelihood, while immunotherapy plus chemotherapy followed by surgery and adjuvant chemotherapy (IC-S-C) demonstrates good safety.

Conclusion S-T and T-S-T treatments have the greatest potential to be the optimal overall survival treatments for stage IIIA-N2 NSCLC patients with positive driver genes, demonstrating significant clinical application prospects. While for patients with negative driver genes, C-S-C treatments benefit the most.

The protocol was registered in the Prospective Register of Systematic Reviews, PROSPERO (CRD42022372711).

Keywords Non-small cell lung cancer, IIIA-N2, Network meta-analysis, Overall survival

Background

The advanced non-small cell lung cancer (NSCLC) is the leading cause of cancer mortality globally [1]. Patients with resectable stage IIIA-N2 disease constitute a significant subset of advanced NSCLC patients, accounting for more than 30% of patients at the time of diagnosis [2, 3]. Although the development of targeted therapy, immunotherapy, and surgery technology has made great progress in the treatment of advanced lung cancer, for patients with stage IIIA-N2 disease, the 5-year survival rate is still lower than 20% [4, 5]. Previous studies have found that

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different treatment choices are important factors leading to poor prognosis in patients with resectable stage IIIA-N2 NSCLC [6, 7]. The ongoing debate regarding the optimal treatment for N2-stage disease is a significant factor contributing to the unfavorable prognosis in this subset of patients.

According to the recommendations of the National Comprehensive Cancer Network (NCCN) guidelines, patients should choose personalized treatment strategies based on the differences in the size of the primary tumor and the location of distant lymph node metastasis [8, 9]. For patients with resectable stage IIIA-N2 NSCLC, surgical or perioperative radiotherapy and chemotherapy are common treatment options, but with the continuous deepening of targeted and immunotherapy research, this new treatment method has also demonstrated good efficacy [10]. There is a significant disparity in efficacy among these options. Determining the optimal treatment approach for patients with resectable stage IIIA-N2 NSCLC is a pressing issue that researchers are urgently seeking to address.

Therefore, we applied a network meta-analysis (NMA) by using frequentist methods approach to examine the efficacy and safety of management for resectable stage IIIA-N2 non-small cell lung cancer patients. The management options include surgery, radiotherapy, chemotherapy, targeted therapy, immunotherapy and combinations of multiple chemotherapeutic agents. This study aims to preliminarily compare the differences between various treatment modalities, especially the effects of targeted therapy, providing new insights for determining the optimal treatment approach for patients with resectable stage IIIA-N2 NSCLC.

Materials and methods

Study selection

We searched PubMed, Embase, MEDLINE, Web of Science and Cochrane Central Register of Controlled Trials databases with the key words “non-small cell lung cancer”, “NSCLC”, “N2” and “IIIA”. Articles written in English and published before March 1st 2023 are included. Each management’s relative efficacy and safety was evaluated in terms of overall survival (OS) and treatment-related death (trDeath). The protocol was registered in the Prospective Register of Systematic Reviews, PROSPERO (CRD42022372711). Two review authors independently evaluated the literatures and extracted information from eligible studies. Any disagreement was resolved by discussion, and a third author was consulted to achieve consensus when necessary.

The inclusion criteria were as follows: randomized controlled trials (RCTs); trials that include patients with pathologically confirmed or clinically suspected stage

IIIA-N2 NSCLC; trials that compared the efficacy of managements including surgery, radiotherapy, chemotherapy, targeted molecular therapy, immunotherapy and multiple combinations thereof; trials that included survival data as endpoints; and full-text articles published in English.

The exclusion criteria were as follows: nonrandomized controlled trials; trials exclusively recruited patients with unresectable stage IIIA-N2 NSCLC; studies investigating IIIA-N2 population in sub analyses; and trials that did not include survival data as endpoints.

Our searching strategy is depicted in the Supplemental Material.

Data extraction and quality assessment

We screened the full text and collated information from eligible studies, including the name of first author, year of publication, study period and source, study design, patient information, managements, and outcomes (Table 1). The primary endpoints were OS and median OS for different management strategies, and the secondary endpoint was trDeath. The Cochrane Risk of Bias Tool was used to assess the risk of bias of each RCT [11]. RCTs defined as having a low risk of bias using the tool were considered high quality.

Statistical analysis

Network meta-analyses were performed according to the frequentist framework in Stata software version 17.0 in random-effects mode [34, 35]. The odds ratio (OR) and 95% confidence intervals (95%CI) were generated as effect sizes for binary variants. The median OS of each management group was compared, and the results are presented as a network forest (shown as OR and 95%CI) to show the pairwise comparisons. The ranking of interventions was determined using the surface under the cumulative ranking curve area (SUCRA). PrBest, which means the probability of this treatment plan of being ranked first, was also shown in the results of the network meta-analysis. Forest plots and league tables of the relative management effects were used to visualize comparisons of network estimations. Both gross and loops inconsistency between direct and indirect comparisons were assessed. $P < 0.05$ was considered to indicate statistical significance. To assess publication bias, a network funnel plot was constructed to detect small sample effects. Additionally, a sensitivity analysis was performed by excluding data from nonsurgical trials.

Result

Eligible studies

We retrieved a total of 3659 articles through our initial searching strategies. After scrupulous screening of these

Table 1 Main characteristic of all the studies included in network meta-analysis

Num	First Author (Year)	Sample size	Median age	Male (%)	Study Period	country	Institution	Text	Control	Squamous (%)	N2-single (%)	Lobectomy (%)	Smoker (%)	OS (month)	trDeath	TNM edition
1	Pass (1992) [12]	13/14	51/55	31/57	1987–1991	America	multiple	C-S-C	Etoposide (120 mg/m2) and cis-platinum (80 mg/m2) every 21-days for two cycles, then surgery, and follow by four additional cycles of chemotherapy of the same regimen	S-R	23.1/21.4	54/43	NG	28.7/15.6	0/0	NG
2	Pisters (1994) [13]	36/36	56/59	53/50	1981–1987	America	single	S-CR	Surgery followed by cisplatin (120 mg/m2 or 3 mg/kg, whichever was less) on days 1, 29, 71 and 113 and vin-desine (3 mg/m2) on days 1, 8, 15, 22, 29, and concurrent radiotherapy (40 Gy)	S-R	19.4/25	39/39	NG	16.3/19.1	1/0	NG
3	Roth (1994) [14]	28/32	50/59	68/78	1987–1993	America	multiple	C-S-C	Cytophosphamide (500 mg/m2) on day 1, etoposide (100 mg/m2) on days 1, 2, 3, and cisplatin (100 mg/m2) on day 1 for three cycles followed by surgery, then three additional cycles of chemotherapy of the same regimen	S	39.3/34.4	NG	NG	64/11	1/2	1986 classification, ISLIC
4	Debevec (1996) [15]	35/39	59/59	86/79	1988–1992	Slovenia	single	S-R	Surgery followed by radiotherapy (2.5–3 Gy daily, 5 or 6 days/week to 30 Gy)	S	34.3/51.3	40/48	NG	28.2/20.2	0/0	NG
5	Shepherd (1998) [16]	16/15	61/52	75/67	NG	Canada	single	C-S	Cisplatin (120 mg/m2) on days 1 and 29 and vin-blastine (6 mg/m2) on days 1, 15, 22, 29 and 43	R	43.8/33.3	NG	NG	18.7/16.2	0/0	NG
6	Rosell (1999) [17]	30/30	60/63	97/100	1989–1996	Spain	multiple	C-S-R	Surgery followed by cisplatin (120 mg/m2 or 3 mg/kg, whichever was less) on days 1, 29, 71 and 113 and vin-desine (3 mg/m2) on days 1, 8, 15, 22, 29, and concurrent radiotherapy (40 Gy)	S-R	63/76.7	27/27	NG	22/10	2/2	1986 classification, ISLIC

Table 1 (continued)

Num	First Author (Year)	Sample size	Median age	Male (%)	Study Period	country	Institution	Text	Control	Squamous (%)	N2-single (%)	Lobectomy (%)	Smoker (%)	OS (month)	trDeath	TNM edition
7	Nagai (2003) [18]	31/31	59/61	65/68	1993–1998	Japan	single	C-S Cisplatin (80 mg/m2) on day 1 and vindesine (3 mg/m2) on days 1 and 8, every 4 weeks for three cycles followed by surgery	S Surgery	22.6/25.8	NG	68/77	NG	17/16	0/0	NG
8	Tada (2004) [19]	59/60	62/62	68/62	1994–1998	Japan	multiple	S-C Surgery followed by bycisplatin (80 mg/m2) on day 1 and vindesine (3 mg/m2) on days 1 and 8 every 4 weeks for three cycles	S Surgery	15.3/25.0	52/47	90/90	NG	36/36	0/0	1997 classification, ISLCC
9	Van Meerbeeck(2007) [20]	167/165	62/61	71/77	1994–2002	Belgium	multiple	C-S Cisplatin (≥80 mg/m2) or carboplatin (AUC≥5) combined with at least one other regimen for three cycles followed by surgery	C-R Cisplatin (≥80 mg/m2) or carboplatin (AUC≥5) with at least one other regimen for three cycles followed by radiotherapy (60–62.5 Gy to the involved mediastinum, or 40–46 Gy to the uninvolved mediastinum)	38.9/40.0	NG	38/NG	NG	16.4/17.5	6/1	1997 classification, ISLCC
10	Perry (2007) [21]	19/18	66/61	63/72	1998–2000	America	single	S-C-R Surgery followed by paclitaxel (200 mg/m2) and carboplatin (AUC=6) every 3 weeks for four cycles and then radiotherapy (50 Gy in 25 fractions over 5 weeks)	S-C Surgery followed by paclitaxel (200 mg/m2) and carboplatin (AUC=6) every 3 weeks for four cycles	NG	NG	68/72	NG	41.5/33.2	0/0	1986 classification, ISLCC
11	Albain (2009) [22]	202/194	59/61	64.9/62.1	1994–2001	America+Canada	multiple	CR-S cisplatin (50 mg/m2 days 1, 8, 29, 36) and etoposide (50 mg/m2 days 1–5 and 29–33), plus 45 Gy thoracic RT beginning day 1, in 1.8 Gy daily fractions, followed by surgery	CR cisplatin (50 mg/m2 days 1, 8, 29, 36) and etoposide (50 mg/m2 days 1–5 and 29–33), plus 61 Gy thoracic RT beginning day 1, in 1.8 Gy daily fractions	31.7/33.5	75.3/63	63/NG	NG	23.6/22.2	16/4	1988 classification, ISLCC
12	Ou (2010) [23]	79/71	54/59	70.8/76.1	1999–2003	Chian	single	S-C Surgery followed by vinorelbine (25 mg/m2) and carboplatin (AUC=5) on days 1 and 8, or paclitaxel (175 mg/m2) and carboplatin (AUC=5) on day 1 for four cycles	S Surgery	22.8/33.8	63.3/77.5	86.1/76.1	NG	33/24	0/0	1997 classification, ISLCC

Table 1 (continued)

Num	First Author (Year)	Sample size	Median age	Male (%)	Study Period	country	Institution	Text	Control	Squamous (%)	N2-single (%)	Lobectomy (%)	Smoker (%)	OS (month)	trDeath	TNM edition
13	Katakami (2012) [24]	31/29	58/57	68/66	2000–2005	Japan	multiple	CR-S	Carboplatin (AUC = 5) and docetaxel (60 mg/m ²) on days 1, 1, 22 for two cycles and concurrent radiotherapy (2 Gy in 20 fractions to 40 Gy over 4 weeks) followed by surgery	C-S	Carboplatin (60 mg/m ²) on days 1, 22 for two cycles followed by surgery	79/69	74/76	39.6/29.9	0/0	6th edition, AJCC
14	Shen (2014) [25]	60/69	58/60	65/65	2004–2009	China	multiple	S-CR	Surgery followed by cisplatin (60 mg/m ²) and paclitaxel (175 mg/m ²) every 3 weeks for four cycles and concurrent radiotherapy (1.8 Gy in 28 fractions to 50.4 Gy)	S-C	Surgery followed by cisplatin (60 mg/m ²) and paclitaxel (175 mg/m ²) every 3 weeks for four cycles	73/72	NG	40/28	1/0	6th edition, AJCC
15	Li (2014) [26]	30/30	59.5/54.6	56.7/60	2008–2011	China	single	S-C-T	Surgery followed by pemetrexed 600 mg/m ² plus carboplatin (AUC = 5) every 3 weeks for four cycles and then gefitinib 250 mg/day orally for 6 months	S-C	Surgery followed by pemetrexed (500 mg/m ²) plus carboplatin (AUC = 5) every 3 weeks for four cycles	93.3/96.7	43.3/46.7	41.6/32.6	0/0	1997 classification, ESIC
16	Pless (2015) [27]	117/115	60/59	67/67	2001–2012	Switzerland	multiple	C-R-S	Cisplatin (100 mg/m ²) and docetaxel (85 mg/m ²) every 3 weeks for three cycles followed by radiotherapy (44 Gy in 22 fractions over 3 weeks) and then surgery	C-S	Cisplatin (100 mg/m ²) and docetaxel (85 mg/m ²) every 3 weeks for three cycles followed by surgery	59/63	91/96	37.1/26.2	0/3	6th edition, AJCC
17	Sun (2017) [28]	51/50	60/60	52/55	2009–2014	Korea	single	S-CR	Surgery followed by concurrent radiotherapy (50 Gy in 25 fractions) with weekly paclitaxel (50 mg/m ²) and cisplatin (25 mg/m ²) for five cycles, then paclitaxel (175 mg/m ²) and cisplatin (80 mg/m ²) every 3 weeks for two cycles	S-C	Surgery followed by paclitaxel (175 mg/m ²) and carboplatin (AUC 5.5) every 3 weeks for four cycles	82/88	55/50	74.3/83.5	0/0	7th edition, AJCC
18	Hui (2021) [29]	184/180	55/55	58.7/	2009–2017	China	single	S-CR	Surgery followed by chemotherapy of platinum-based doublet regimen for four cycles and then radiotherapy (2 Gy per fraction up to 50 Gy over 5 weeks)	S-C	Surgery followed by chemotherapy of platinum-based doublet regimen for four cycles	NG	44.5/48.9	NR/81.5	0/0	6th before Jan 2010, 7th thereafter

Table 1 (continued)

Num	First Author (Year)	Sample size	Median age	Male (%)	Study Period	country	Institution	Text	Control	Squamous (%)	N2-single (%)	Lobectomy (%)	Smoker (%)	OS (month)	trDeath	TNM edition
19	Pechoux (2022) [30]	252/249	61/61	66/66	2007–2018	Europe	multiple	S-C-R	Surgery followed by chemotherapy of platinum-based doublet regimen for four cycles, and then radiotherapy(3D conformal radiotherapy 54 Gy in 27 fractions of 2.0 Gy or 30 fractions of 1.8 Gy)	S-C	Surgery followed by chemotherapy of platinum-based doublet regimen for four cycles	78/81	92/89	NR/NR	3/0	7th edition, AJCC
20	Forde (2022) [31]	113/115	64/65	71.5/70.9	2017–2019	America	multiple	IC-S-C	nivolumab (360 mg) plus platinum-doublet chemotherapy every 3 weeks for three cycles followed by surgery, and then platinum-doublet chemotherapy every 3 weeks for three cycles	C-S-C	platinum-doublet chemotherapy every 3 weeks for three cycles followed by surgery, and then the same chemotherapy regimen for three cycles	NG	89.4/88.3	NR/NR	0/3	7th edition, AJCC
21	Yue (2022) [32]	51/51	59/57	17/20	2012–2015	China	Multiple	S-T	Surgery followed by erlotinib 150 mg/d orally for 2 years	S-C	Surgery followed by vinorelbine 25 mg/m2(days 1 and 8, intravenous) plus cisplatin 75 mg/m2 (day 1) every 3 weeks for four cycles	96/98	25/24	84.2/61.1	0/0	7th edition, AJCC
22	Zhong (2023) [33]	37/35	59/58	29.7/22.9	2011–2017	China	multiple	T-S-T	erlotinib 150 mg/d orally for 42 days, followed by surgery, and then erlotinib 150 mg/d orally for 12 months	C-S-C	Gemcitabine 1,250 mg/m2 (days 1 and 8, intravenous) plus cisplatin 75 mg/m2 (day 1) every 3 weeks for two cycles followed by surgery, and then the same chemotherapy regimens for two cycles	64.9/54.3	21.6/11.4	42.2/36.9	0/0	NG

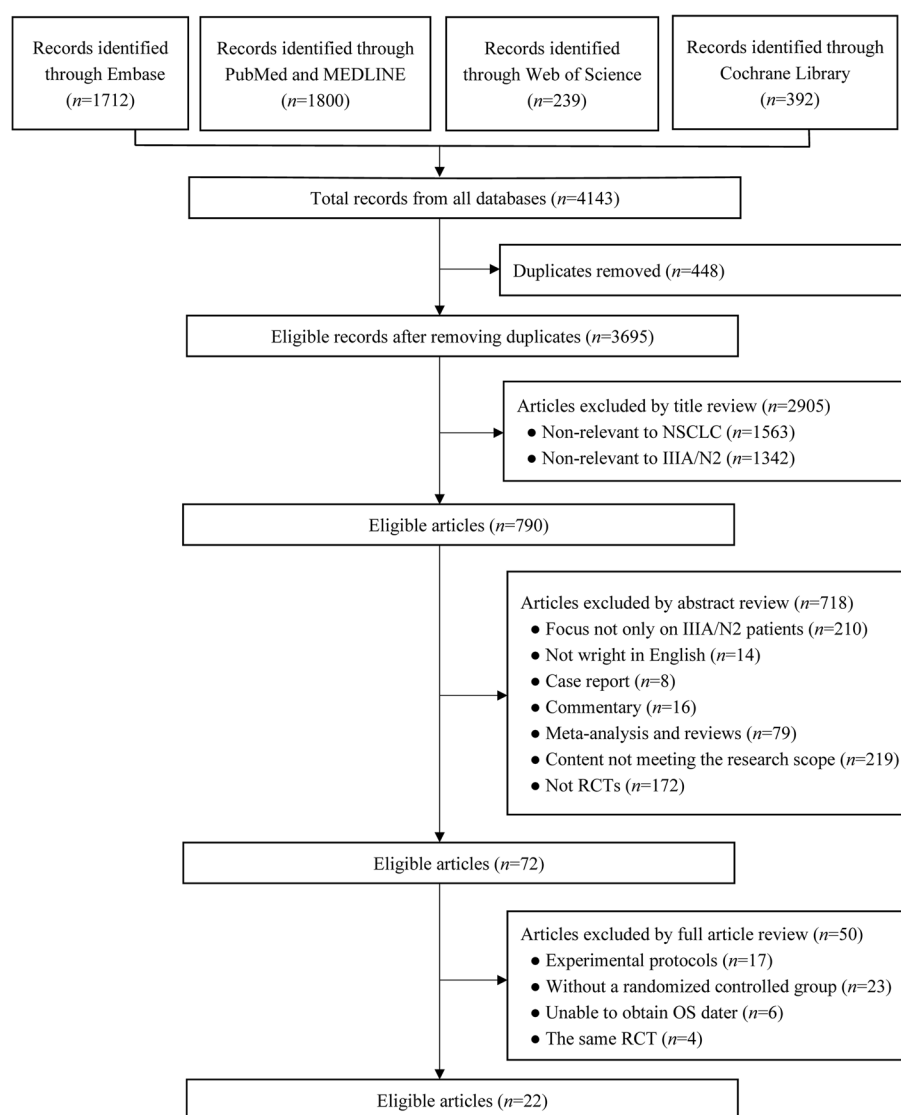


Fig. 1 Flow chart of articles retrieved, screened and included in this network meta-analysis

articles, a total of 22 RCTs [12–33] with 3269 participants were included. Figure 1 shows the process of study selection. All 22 studies were of high quality according to the Cochrane Risk of Bias Tool of RCTs (Supplement Fig. 1 and Supplement Table 1).

The participants of the RCTs received 1 of the following 17 managements (Fig. 2): radiotherapy alone (R), surgery alone (S), concurrent chemoradiotherapy (CR), sequential chemoradiotherapy (C-R), neoadjuvant chemotherapy followed by surgery (C-S), neoadjuvant concurrent chemoradiotherapy followed by surgery (CR-S), surgery followed by adjuvant chemotherapy (S-C), surgery followed by adjuvant radiotherapy (S-R), surgery followed by adjuvant targeted therapy (S-T), surgery followed by

adjuvant concurrent chemoradiotherapy (S-CR), surgery followed by adjuvant sequential chemoradiotherapy (S-C-R), surgery followed by adjuvant chemotherapy and adjuvant targeted therapy (S-C-T), neoadjuvant sequential chemoradiotherapy followed by surgery (C-R-S), neoadjuvant chemotherapy followed by surgery and adjuvant chemotherapy (C-S-C), neoadjuvant chemotherapy followed by surgery and adjuvant radiotherapy (C-S-R), neoadjuvant concurrent immunotherapy plus chemotherapy followed by surgery and adjuvant chemotherapy (IC-S-C), neoadjuvant targeted therapy followed by surgery and adjuvant targeted therapy (T-S-T). All 22 RCTs reported information on both overall survival and trDeath. The outcome of comparisons of overall survival

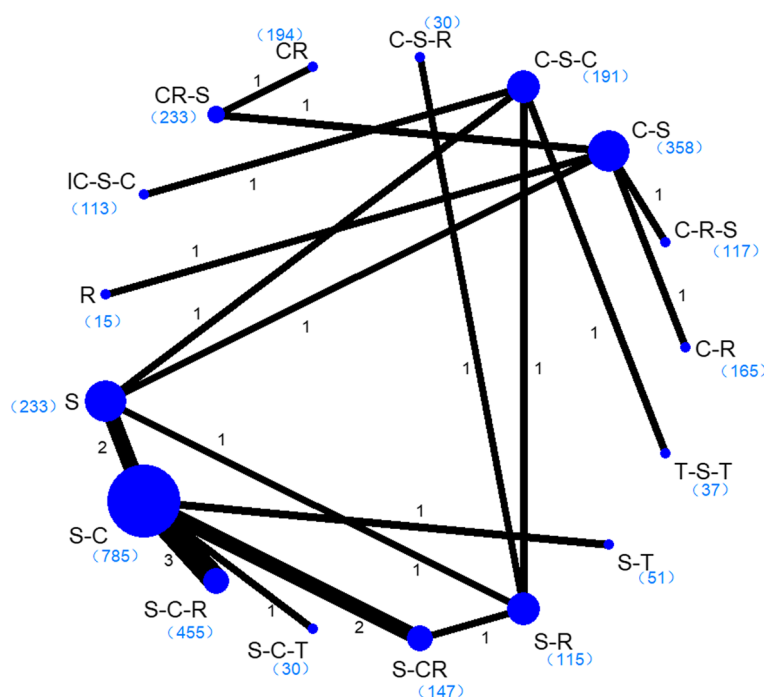


Fig. 2 Network of comparisons established for this network meta-analysis. The blue circle represents the total number of people receiving a certain treatment. The black line connecting the circles represents the number of RCTs comparing different treatment options. R: radiotherapy; S: surgery; CR: concurrent chemoradiotherapy; C-R: sequential chemoradiotherapy; C-S: neoadjuvant chemotherapy + surgery; CR-S: neoadjuvant concurrent chemoradiotherapy + surgery; S-C: surgery + adjuvant chemotherapy; S-R: surgery + adjuvant radiotherapy; S-T: surgery + adjuvant targeted therapy; S-CR: surgery + adjuvant concurrent chemoradiotherapy; S-C-R: surgery + adjuvant sequential chemoradiotherapy, S-C-T: surgery + adjuvant chemotherapy + adjuvant targeted therapy, C-R-S: neoadjuvant sequential chemoradiotherapy + surgery, C-S-C: neoadjuvant chemotherapy + surgery + chemotherapy; C-S-R: neoadjuvant chemotherapy + surgery + adjuvant radiotherapy; IC-S-C: neoadjuvant concurrent immunotherapy plus chemotherapy + surgery + adjuvant chemotherapy, T-S-T: neoadjuvant targeted therapy + surgery + adjuvant targeted therapy

and trDeath between groups in the RCTs are presented in Supplemental Fig. 2A and Supplemental Fig. 2B.

Publication bias

The network funnel plot for overall survival and trDeath is showed in Supplement 3. Most studies can be symmetrically distributed on both sides of the perpendicular, indicating a lower likelihood of small sample effects.

Inconsistency

The studies included, whether at the overall or subgroup level, did not exhibit significant inconsistency ($p=0.6349$), allowing for the use of a consistency model analysis. A triangular loop (S, C-S-C, S-R) and a quadratic loop (S, S-C, S-CR, S-R) are found in network meta-analyses. Because only 1 RCT provided evidence for each direct comparison, the heterogeneity of the triangular loop could not be estimated. The evaluation of inconsistency of the triangular using loop-specific heterogeneous estimates did not reveal significant inconsistency (IF 0.110, 95% CI 0.00–4.47, $p=0.961$).

Network Meta-analyses for efficacy and safety

Our results from the random-effects network meta-analysis are presented in Table 2.

Compared to surgery or radiotherapy alone, adjuvant and neoadjuvant treatment significantly extends the survival of patients. Among them, C-S-C, S-T, and T-S-T demonstrate favorable outcomes, showing higher survival benefits compared to other treatment modalities. C-S-C exhibits significant survival benefits compared to C-S (OR 0.07, 95% CI 0.02–0.25) and S-C (OR 4.87, 95% CI 1.18–20.12) (Supplement Fig. 3A). Meanwhile, in studies on postoperative radiotherapy, compare with C-S, C-R-S (OR 2.45, 95% CI 0.89–6.79) and CR-S (OR 1.92, 95% CI 0.74–5.50) had no significant survival benefits (Supplement Fig. 3B). Similarly, there was no significant difference between S-C and S-C-R (OR 2.58, 95% CI 0.93–7.17), or between S-C and S-CR (OR 1.63, 95% CI 0.68–3.90) (Supplement Fig. 3C). S-T and T-S-T demonstrate significant benefits compared to most treatment modalities, except for the lack of significance in advantages over C-S-C, C-S-R, and IC-S-C (Fig. 3).

Rank probabilities of treatments

The ranking probabilities of all competing treatments in terms of efficacy and safety are shown in Fig. 4A and Supplemental Fig. 5A. The surface under the cumulative ranking curve (SUCRA) was used to evaluate the merits of the treatment, which suggested that the potential best treatment options for extending overall survival are S-T (SUCRA = 94.6) and T-S-T (SUCRA = 89.7). For overall survival, S-T has the highest probabilities of being ranked first (59.8%), while radiotherapy alone and surgery alone are ranked the lowest (Fig. 4B). In terms of safety, IC-S-C has the highest probability of being ranked first (31%), while S-CR and S-C-R have the highest probability of being ranked last (Supplement Fig. 5B).

We excluded trials conducted on patients with targeted therapy and conducted a second network meta-analysis. Results are showed in Supplement Fig. 6, which suggested that the best option is C-S-C (SUCRA = 94.8), and the worst options is R alone (SUCRA = 15.5).

Sensitivity analysis

A sensitivity analysis using overall survival data was performed, excluding 3 trials [16, 20, 22]. The results are showed in Supplement Fig. 7, which suggested that the best options are S-T (SUCRA = 93.7) and T-S-T (SUCRA = 86.3) and the worst option was surgery alone (SUCRA = 86.3). The results are consistent with the native analysis and confirm the robustness and reliability of the overall conclusions.

Discussion

To our knowledge, this study is currently the first NMA targeting patients with stage IIIA-N2 NSCLC and includes data on targeted therapy and immunotherapy. It suggests that S-T and T-S-T probably provide the best OS adjuvant treatments might be selected depending on the heterogeneity analysis of the tumor involvement of mediastinal lymph nodes by a multidisciplinary team. In the studies included, most safety comparisons showed no differences, primarily because of good safety profiles, with most experiments not resulting in trDeath. This lends further support to the hypothesis that stage IIIA-N2 NSCLC is potentially a systemic disease, emphasizing the crucial role of systemic therapy in managing both local regions and potential metastases.

Lung cancer is a significant health concern and the treatment of stage IIIA-N2 NSCLC remains a topic of controversy within the medical community [36]. The controversies primarily revolve around the effectiveness and optimal approach to treating this specific stage of lung cancer, including surgery(S), radiotherapy(R) and chemotherapy(C), along with various treatment modalities [37–39]. The research results indicate that patients

who received combined systemic treatment tend to achieve greater survival benefits than those who undergo either radiotherapy or surgery alone. The incorporation of neoadjuvant chemotherapy holds the potential for increased chances of tumor down-staging, facilitating comprehensive surgery, and eliminating microresidual disease, which can ultimately translate into a survival benefit. Our results indicate that, in the absence of targeted therapy, C-S-C is the most effective treatment option. In comparison with C-S, postoperative adjuvant chemotherapy demonstrated significant benefits. Additionally, compared to C-S-R, C-S-C exhibited greater efficacy and higher safety. The ongoing discussion about the potential advantages of induction radiation therapy is reflected in our findings. Neither C-R-S nor CR-S showed a significant survival benefit over C-S. Furthermore, in the comparison between S-C-R and S-C, S-C-R didn't demonstrate a significant survival benefit over S-C. This suggests that in the treatment of IIIA-N2 NSCLC, neoadjuvant and adjuvant concurrent radiotherapy for the general population does not significantly improve survival. This conclusion aligns with those of some recently published retrospective studies [40–42]. Therefore, routine radiotherapy is not recommended. The role and appropriate patient population for perioperative radiotherapy in the treatment of IIIA-N2 NSCLC still require further research [43, 44].

In recent years, significant advancements in targeted therapy and immunotherapy have changed the treatment pattern of NSCLC, and also have brought about new treatment options and insights for stage IIIA-N2 NSCLC [45–47]. So we do the first NMA targeting patients with stage IIIA-N2 NSCLC and includes data on targeted therapy and immunotherapy.

Our study revealed that S-T and T-S-T may be the optimal treatment regimens for prolonging survival, and they exhibit considerable safety. This finding, presented for the first time in this paper, aligns with the results of many retrospective studies. Targeted therapy has demonstrated outstanding efficacy in patients with stage IIIA-N2 NSCLC. The ADAURA study [48], ADJUVANT study [49], and EVAN study [50] have demonstrated significant progression-free survival (PFS) benefits with adjuvant targeted therapy, especially in NSCLC patients with certain EGFR gene mutations, notably in stage IIIA. The EMERGING-CTONG 1103 study [51] has explored beneficial attempts at neoadjuvant and adjuvant targeted therapy in EGFR-positive stage IIIA-N2 patients. However, it is important to note that all targeted therapies included in this study were first-generation TKIs, and the users of these drugs were patients with positive driver genes. For patients with negative driver genes, C-S-C might still be the best treatment option. The effectiveness

Table 2 Estimate results according to the Network Meta-Analysis

C-R	2.25 (0.55,9.30)	0.92 (0.34,2.46)	13.32 (2.62,67.70)	5.71 (0.88,37.25)	1.60 (0.30,8.56)	1.77 (0.45,6.98)	13.32 (0.18,991.01)	0.70 (0.16,3.08)	0.85 (0.21,3.44)	1.06 (0.23,4.82)	2.74 (0.44,17.03)	2.81 (0.43,18.50)	1.73 (0.33,9.02)	2.17 (0.45,10.45)	43.75 (2.71,706.12)	16.46 (2.65,102.12)
0.44 (0.11,1.88)	C-R-S	0.41 (0.15,1.13)	5.92 (1.14,30.65)	2.53 (0.38,16.82)	0.71 (0.13,3.87)	0.78 (0.19,3.17)	5.92 (0.08,443.25)	0.31 (0.07,1.40)	0.38 (0.09,1.56)	0.47 (0.10,2.18)	1.22 (0.19,7.69)	1.25 (0.19,8.35)	0.77 (0.14,4.08)	0.97 (0.20,4.73)	19.43 (1.19,317.11)	7.31 (1.16,46.13)
1.09 (0.41,2.92)	2.45 (0.89,6.79)	C-S	14.50 (3.99,52.78)	6.22 (1.26,30.63)	1.74 (0.45,6.76)	1.92 (0.74,5.00)	14.50 (0.22,962.10)	0.76 (0.25,2.30)	0.93 (0.34,2.49)	1.16 (0.37,3.63)	2.98 (0.64,13.88)	3.06 (0.61,15.24)	1.88 (0.50,7.08)	2.37 (0.70,8.02)	47.63 (3.54,641.45)	17.92 (3.86,83.20)
0.08 (0.01,0.38)	0.17 (0.03,0.88)	0.07 (0.02,0.25)	C-S-C	0.43 (0.11,1.62)	0.12 (0.02,0.78)	0.13 (0.03,0.66)	1.00 (0.02,54.10)	0.05 (0.01,0.29)	0.06 (0.03,0.15)	0.08 (0.03,0.21)	0.21 (0.05,0.85)	0.21 (0.05,0.94)	0.13 (0.04,0.38)	0.16 (0.07,0.38)	3.28 (0.26,41.25)	1.24 (0.54,2.83)
0.18 (0.03,1.14)	0.39 (0.06,2.62)	0.16 (0.03,0.79)	2.33 (0.62,8.80)	C-S-R	0.28 (0.03,2.27)	0.31 (0.05,1.99)	2.33 (0.03,156.52)	0.12 (0.02,0.85)	0.15 (0.04,0.52)	0.19 (0.05,0.69)	0.48 (0.09,2.55)	0.49 (0.09,2.78)	0.30 (0.08,1.12)	0.38 (0.14,1.06)	7.66 (0.53,111.87)	2.88 (0.60,13.79)
0.62 (0.12,3.34)	1.41 (0.67,7.66)	0.57 (0.15,2.22)	8.32 (1.28,54.09)	3.57 (0.44,28.91)	CR	1.10 (0.42,2.89)	8.32 (0.10,683.27)	0.43 (0.08,2.51)	0.53 (0.10,2.84)	0.66 (0.11,3.91)	1.71 (0.22,13.28)	1.75 (0.21,14.34)	1.08 (0.16,7.18)	1.36 (0.22,8.41)	27.33 (1.46,512.82)	10.28 (1.33,79.67)
0.57 (0.14,2.23)	1.27 (0.32,5.15)	0.52 (0.20,1.35)	7.54 (1.51,37.57)	3.23 (0.50,20.73)	0.91 (0.35,2.37)	CR-S	7.54 (0.10,556.67)	0.39 (0.09,1.71)	0.48 (0.12,1.90)	0.60 (0.14,2.67)	1.55 (0.25,9.47)	1.59 (0.25,10.29)	0.98 (0.19,5.01)	1.23 (0.26,5.80)	24.75 (1.55,395.07)	9.31 (1.53,56.80)
0.08 (0.00,5.58)	0.17 (0.00,12.67)	0.07 (0.00,4.57)	1.00 (0.02,54.10)	0.43 (0.01,28.74)	0.12 (0.00,9.87)	0.13 (0.00,9.80)	IC-S-C	0.05 (0.00,4.00)	0.06 (0.00,3.76)	0.08 (0.00,4.85)	0.21 (0.00,14.21)	0.21 (0.00,14.93)	0.13 (0.00,8.08)	0.16 (0.00,9.64)	3.28 (0.03,370.44)	1.24 (0.02,72.81)
1.44 (0.32,6.37)	3.24 (0.72,14.65)	1.32 (0.43,4.02)	19.16 (3.48,105.43)	8.21 (1.17,57.42)	2.30 (0.40,13.30)	2.54 (0.59,11.02)	19.16 (0.25,1469.68)	R	1.22 (0.28,5.42)	1.53 (0.31,7.54)	3.94 (0.59,26.30)	4.04 (0.57,28.51)	2.49 (0.44,14.03)	3.13 (0.60,16.32)	62.93 (3.72,1064.74)	23.68 (3.55,157.72)
1.18 (0.29,4.76)	2.65 (0.64,10.97)	1.08 (0.40,2.91)	15.68 (6.84,35.96)	6.72 (1.92,23.47)	1.88 (0.35,10.09)	2.08 (0.53,8.23)	15.68 (0.27,923.91)	0.82 (0.18,3.63)	S	1.25 (0.70,2.22)	3.22 (0.99,10.46)	3.31 (0.93,11.71)	2.03 (0.84,4.91)	2.56 (1.25,5.23)	51.49 (4.65,570.19)	19.37 (5.99,62.65)
0.94 (0.21,4.26)	2.12 (0.46,9.81)	0.86 (0.28,2.71)	12.54 (4.72,33.27)	5.37 (1.44,20.05)	1.51 (0.26,8.88)	1.66 (0.37,7.38)	12.54 (0.21,762.92)	0.65 (0.13,3.23)	0.80 (0.45,1.42)	S-C	2.58 (0.93,7.17)	2.64 (0.86,8.15)	1.63 (0.68,3.90)	2.05 (0.90,4.67)	41.17 (3.99,425.20)	15.49 (4.30,55.77)
0.37 (0.06,2.27)	0.82 (0.13,5.20)	0.34 (0.07,1.56)	4.87 (1.18,20.12)	2.08 (0.39,11.10)	0.58 (0.08,4.54)	0.65 (0.11,3.95)	4.87 (0.07,336.28)	0.25 (0.04,1.70)	0.31 (0.10,1.01)	0.39 (0.14,1.08)	S-C-R	1.03 (0.22,4.70)	0.63 (0.16,2.46)	0.79 (0.21,2.97)	15.98 (1.25,204.46)	6.01 (1.16,31.12)
0.36 (0.05,2.34)	0.80 (0.12,5.37)	0.33 (0.07,1.63)	4.74 (1.07,21.05)	2.03 (0.36,11.50)	0.57 (0.07,4.66)	0.63 (0.10,4.08)	4.74 (0.07,335.87)	0.25 (0.04,1.75)	0.30 (0.09,1.07)	0.38 (0.12,1.17)	0.97 (0.21,4.46)	S-C-T	0.62 (0.15,2.56)	0.77 (0.19,3.13)	15.58 (1.17,208.11)	5.86 (1.06,32.27)
0.58 (0.11,3.02)	1.30 (0.24,6.94)	0.53 (0.14,2.00)	7.71 (2.65,22.46)	3.30 (0.89,12.24)	0.93 (0.14,6.17)	1.02 (0.20,5.24)	7.71 (0.12,480.18)	0.40 (0.07,2.27)	0.49 (0.20,1.19)	0.62 (0.26,1.48)	1.58 (0.41,6.17)	1.63 (0.39,6.77)	S-CR	1.26 (0.56,2.84)	25.32 (2.09,306.55)	9.53 (2.46,36.87)
0.46 (0.10,2.21)	1.04 (0.21,5.08)	0.42 (0.12,1.43)	6.13 (2.64,14.21)	2.63 (0.94,7.33)	0.74 (0.12,4.56)	0.81 (0.17,3.83)	6.13 (0.10,361.92)	0.32 (0.06,1.67)	0.39 (0.19,0.80)	0.49 (0.21,1.12)	1.26 (0.34,4.72)	1.29 (0.32,5.22)	0.79 (0.35,1.79)	S-R	20.12 (1.69,239.49)	7.57 (2.32,24.68)
0.02 (0.00,0.37)	0.05 (0.00,0.84)	0.02 (0.00,0.28)	0.30 (0.02,3.83)	0.13 (0.01,1.90)	0.04 (0.00,0.69)	0.04 (0.00,0.64)	0.30 (0.00,34.35)	0.02 (0.00,0.27)	0.02 (0.00,0.22)	0.02 (0.00,0.25)	0.06 (0.00,0.80)	0.06 (0.00,0.86)	0.04 (0.00,0.48)	0.05 (0.00,0.59)	S-T	0.38 (0.03,5.40)
0.06 (0.01,0.38)	0.14 (0.02,0.86)	0.06 (0.01,0.26)	0.81 (0.35,1.85)	0.35 (0.07,1.66)	0.10 (0.01,0.75)	0.11 (0.02,0.65)	0.81 (0.01,47.68)	0.04 (0.01,0.28)	0.05 (0.02,0.17)	0.06 (0.02,0.23)	0.17 (0.03,0.86)	0.17 (0.03,0.94)	0.10 (0.03,0.41)	0.13 (0.04,0.43)	2.66 (0.19,38.11)	T-S-T

The odds ratio (OR) and 95% confidence intervals (95%CI) were generated as effect sizes for binary variants. The numerical value highlighted in red indicates that the comparison between the two vehicles has statistical significance

R radiotherapy, S surgery, CR concurrent chemoradiotherapy, C-R sequential chemoradiotherapy, C-S neoadjuvant chemotherapy+surgery, CR-S neoadjuvant concurrent chemoradiotherapy+surgery, S-C surgery+adjuvant chemotherapy, S-T surgery+adjuvant targeted therapy, S-CR surgery+adjuvant concurrent chemoradiotherapy, S-C-R surgery+adjuvant sequential chemoradiotherapy, S-C-T surgery+adjuvant chemotherapy+adjuvant targeted therapy, C-R-S neoadjuvant sequential chemoradiotherapy+surgery, C-S-C neoadjuvant chemotherapy+surgery+chemotherapy, C-S-R neoadjuvant chemotherapy+surgery+adjuvant radiotherapy, IC-S-C neoadjuvant concurrent immunotherapy plus chemotherapy+surgery+adjuvant chemotherapy, T-S-T neoadjuvant targeted therapy+surgery+adjuvant targeted therapy

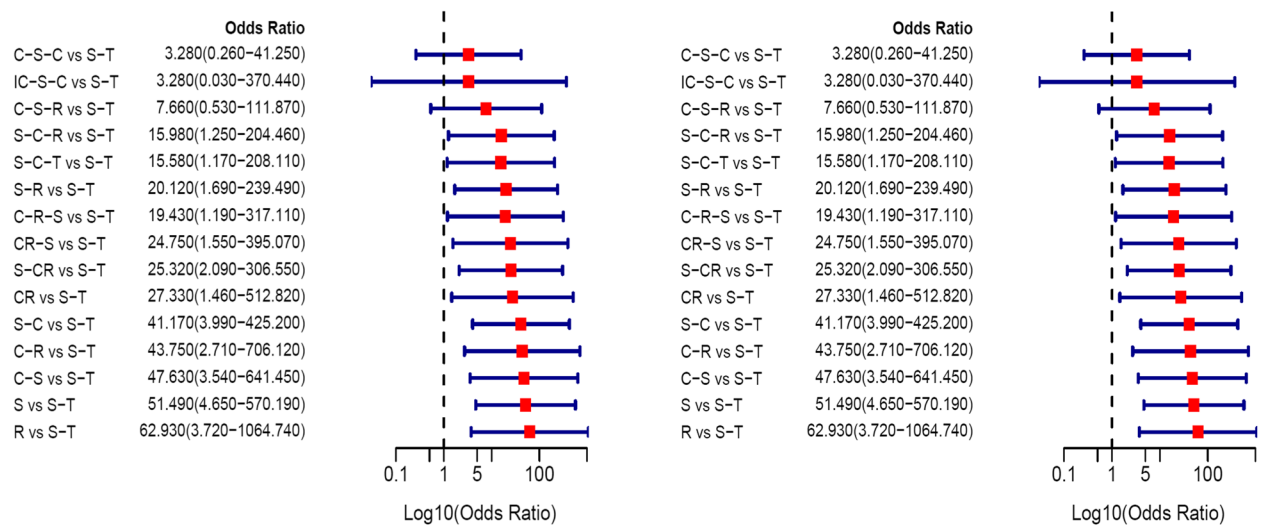


Fig. 3 Forest plot of comparison of management among different groups. R: radiotherapy; S: surgery; CR: concurrent chemoradiotherapy; C-R: sequential chemoradiotherapy; C-S: neoadjuvant chemotherapy + surgery; CR-S: neoadjuvant concurrent chemoradiotherapy + surgery; S-C: surgery + adjuvant chemotherapy; S-R: surgery + adjuvant radiotherapy; S-T: surgery + adjuvant targeted therapy; S-CR: surgery + adjuvant concurrent chemoradiotherapy; S-C-R: surgery + adjuvant sequential chemoradiotherapy, S-C-T: surgery + adjuvant chemotherapy + adjuvant targeted therapy, C-R-S: neoadjuvant sequential chemoradiotherapy + surgery, C-S-C: neoadjuvant chemotherapy + surgery + chemotherapy; C-S-R: neoadjuvant chemotherapy + surgery + adjuvant radiotherapy; IC-S-C: neoadjuvant concurrent immunotherapy plus chemotherapy + surgery + adjuvant chemotherapy, T-S-T: neoadjuvant targeted therapy + surgery + adjuvant targeted therapy

of targeted therapy in these cases requires further research for confirmation.

Advances in immunotherapy are gradually being applied in clinical treatment. The PACIFIC study [7] has confirmed that administering PD-L1 inhibitors after concurrent chemoradiotherapy for unresectable stage III NSCLC can improve OS, a finding acknowledged by major guidelines. This study indicate that IC-S-C has shown promising efficacy, ranking just below S-T and T-S-T, as well as C-S-C, in the assessment of the best treatment options. Moreover, it exhibits the best treatment safety. However, these results are based on a small number of studies, and there is still a considerable amount of undisclosed clinical trial data. The practical value of this treatment approach requires further evaluation. Research into adjuvant immunotherapy, maintenance targeted therapy after concurrent chemoradiotherapy, and other methods is also underway. However, the current landscape of treatment for stage IIIA-N2 patients has not been substantially altered by targeted therapy and immunotherapy studies due to immature overall survival data or a lack of results from large phase III clinical trials.

Limitations

Although NMA has become a valuable tool for synthesizing evidence from multiple sources, it still has many limitations. The date was correctly recorded in this NMA,

and all the included studies were of high quality according to the Cochrane Library. However, estimates of treatment effects may be inaccurate for the following reasons: (1) most treatments were compared indirectly rather than directly; (2) patients with stage IIIA-N2 NSCLC exhibit significant heterogeneity, and differences in tumor resection scope and lymph node dissection varied across studies, make it challenging to analyze different subtypes; (3) we attempted further analysis for disease-free survival (DFS), but the included studies had notable differences in endpoint events, making this part of the research difficult; (4) for most of the trials on perioperative ICI set EFS or DFS as study endpoints and did not publish OS data, or OS data is not yet mature, during the course of this study and using the search function of this study, only one study on perioperative ICI for stage IIIA-N2 lung cancer was included in the study, which may lead to insufficient evaluation of the efficacy of immunotherapy; and (5) the included studies spanned a long period, showing substantial progress in treatment options. This improvement in recent studies resulted in markedly better patient outcomes than in earlier research. Meanwhile, similar to traditional meta-analyses, NMA is susceptible to publication bias. Studies with statistically significant results are more likely to be published, leading to an overestimation of treatment effects. The quality of NMA also depends on the availability and quality of the data. In this study, due to ongoing research on immunotherapy, there

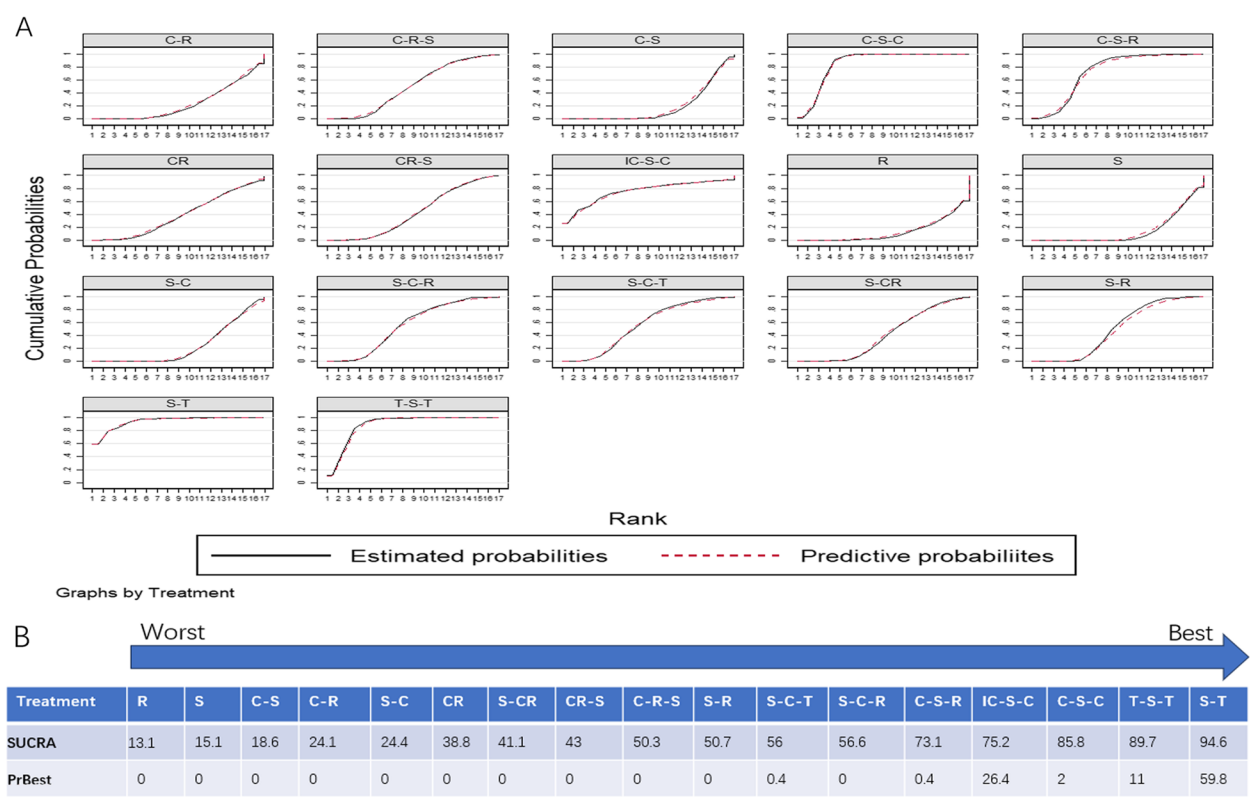


Fig. 4 Ranking curves on efficacy of management according to the results of this network meta-analysis. R: radiotherapy; S: surgery; CR: concurrent chemoradiotherapy; C-R: sequential chemoradiotherapy; C-S: neoadjuvant chemotherapy + surgery; CR-S: neoadjuvant concurrent chemoradiotherapy + surgery; S-C: surgery + adjuvant chemotherapy; S-R: surgery + adjuvant radiotherapy; S-T: surgery + adjuvant targeted therapy; S-CR: surgery + adjuvant concurrent chemoradiotherapy; S-C-R: surgery + adjuvant sequential chemoradiotherapy, S-C-T: surgery + adjuvant chemotherapy + adjuvant targeted therapy, C-R-S: neoadjuvant sequential chemoradiotherapy + surgery, C-S-C: neoadjuvant chemotherapy + surgery + chemotherapy; C-S-R: neoadjuvant chemotherapy + surgery + adjuvant radiotherapy; IC-S-C: neoadjuvant concurrent immunotherapy plus chemotherapy + surgery + adjuvant chemotherapy, T-S-T: neoadjuvant targeted therapy + surgery + adjuvant targeted therapy

are limited publicly disclosed data, potentially introducing bias to this aspect of the results. While these biases could cause deviations in the results, their interpretation remains challenging.

Conclusion

Although the optimal treatment for patients with stage IIIA-N2 NSCLC remains a focal point for subsequent research, this NMA reveals that for patients with positive driver genes, T-S-T and S-T demonstrate significant potential, possibly surpassing the traditional C-S-C approach and offering substantial survival benefits for patients. While for patients with negative driver genes, C-S-C treatments benefit the most. Subsequent disclosure of results from immunotherapy studies may further alter the treatment landscape for this patient subgroup. More high-quality clinical trials are needed to support our findings.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12885-024-13047-2>.

Supplementary Material 1.

Authors' contributions

Qiduo Yu, Data curation; formal analysis; investigation; writing - original draft; writing review & editing; visualization. Haoshuai Yang, Data curation; formal analysis; Investigation; writing-review & editing.Fei Xiao, Data curation; formal analysis; Investigation; writing-review & editing.Zihan Wang, Data curation; formal analysis; Investigation. Zhenrong Zhang, Data curation; formal analysis; Investigation. Qianli Ma, Data curation; formal analysis; Investigation; project administration. Hongxiang Feng, Data curation; formal analysis; Investigation. Zhoujunyi Tian, Data curation; formal analysis; Investigation. Jin Zhang, Data curation; formal analysis; Investigation; project administration. Chaoyang Liang, Conceptualization; methodology; Investigation; project administration; supervision; writing-review & editing; validation.

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Data availability

The data utilized in this meta-analysis are sourced from publicly available published literature. All referenced literature used in this study is accessible through their respective publication channels. We have cited all utilized literature within the manuscript.

Declarations

Ethics approval and consent to participate

Human Ethics and Consent to Participate declarations: not applicable.

Consent for publication

Consent for publication by all authors.

Competing interests

The authors declare no competing interests.

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